Determine whether minimal all-time regret scales with number of agent types

ChatGPT (gpt-5.6-sol)*

Solution generated on 2026-08-27

Verifier verdict: APPROVED — progress score 3/3 (full solution)
Independent A.8 verifier assessment; pipeline details in the attribution footnote.

Abstract

We give a complete solution to the question of whether the optimal anytime regret of a stochastic dynamic matching instance must, or can, grow with the number n of types. Network size alone does not force regret growth on every instance: for every $n \geq 2$, a connected n-type star with unit rewards admits a policy with zero regret at every horizon. In the worst case, however, linear growth is unavoidable. For every $k \geq 2$, we construct a connected simple instance G_k with $n = 3k$ types, rewards in a fixed interval independent of k, and arrival probabilities between $3/(5n)$ and $9/(5n)$, such that

$$\frac{29e^{-9/5}}{6000}\, n \leq \mathcal{R}^*(G_k) \leq \frac{2+\delta}{3}\, n, \qquad \delta = \frac{3e^{-9/5}}{500}.$$

Thus $\mathcal{R}^*(G_k) = \Theta(n)$. The mean matching linear program of G_k has a unique nondegenerate optimum whose minimum positive basic coordinate is $\epsilon = 3/(5n)$.

This completely resolves the stated raw network-size question: growth is not universal over instances, but uniform worst-case guarantees over the class considered here cannot be $o(n)$. The construction gives $\Theta(n) = \Theta(\epsilon^{-1})$ regret. It does not establish an $\Omega(n/\epsilon)$ lower bound and therefore does not prove that a separate multiplicative factor n in an $O(n/\epsilon)$ upper bound is tight.

1 The open problem

This paper addresses an open problem posed by Süleyman Kerimov, Itai Ashlagi, Itai Gurvich in *On the Optimality of Greedy Policies in Dynamic Matching* [1] (Economics and Computation (EC), 2022), Page 17 / 23, Remark 6.2 (Section 6, near end). The website catalogue entry is problem 27 (W4285396203_p0) of the Open Problems in Operations Research collection.

*Generated by the base_solver_non_interactive (two-iteration) pipeline (Codex CLI, non-interactive; reasoning effort ultra; stages A.6 solve → A.8 verify → A.9 write-up; run run_20260827_214924_101ece51, 2it-audit-approved); two-iteration run, outcome from iteration 1. Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

Let $\mathcal{R}^*(G)$ denote the minimal achievable all-time regret level for a given two-way matching network instance $G = (M, \lambda, r)$, defined by

$$\mathcal{R}^*(G) := \inf_{D\in\Pi} \sup_{t>0} \bigl(R_t^* - R_{D,t}\bigr),$$

where Π is the class of admissible (nonanticipative) matching policies.

Determine whetheror specify conditions under which$\mathcal{R}^*(G)$ must grow as a function of the number of agent types n.

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

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2 Model and statement of the result

A stochastic matching instance consists of a finite simple compatibility graph

$$G = (V, E),$$

a reward $r_e > 0$ for each edge $e \in E$, and an arrival probability vector

$$\lambda = (\lambda_i)_{i\in V}, \qquad \lambda_i > 0, \qquad \sum_{i\in V} \lambda_i = 1.$$

Exactly one agent arrives in each period, and its type is drawn independently from λ. Unmatched agents may wait indefinitely. Whenever two currently unmatched agents have types that form an edge of G, a policy may match them and collect the reward of that edge. Matches are irrevocable.

A policy D, possibly randomized, is *nonanticipative*: its decisions through period t are measurable with respect to the arrivals through period t and internal randomness independent of the arrival process. We consider *anytime* policies, so the same policy is evaluated at every possible terminal horizon and does not receive a terminal horizon as an input.

For a realized sequence of arrivals, let $\mathrm{OPT}(t)$ be the maximum reward of any offline matching of the agents arriving in periods $1, \dots, t$, and let $W_D(t)$ be the reward earned by D through period t. Define

$$\mathrm{Reg}_D(t) \coloneqq \mathbb{E}[\mathrm{OPT}(t) - W_D(t)]$$

and

$$\mathcal{R}^*(G) \coloneqq \inf_{D} \sup_{t\in\mathbb{N}} \mathrm{Reg}_D(t).$$

The expectation is over both the arrival process and any internal randomization of D. Since the matches selected by D form a feasible matching of the realized agents,

$$\mathrm{OPT}(t) - W_D(t) \geq 0$$

pathwise.

The distinction between universal and worst-case dependence on $n = |V|$ is essential. The following theorem resolves both quantifiers.

Theorem 1 (Complete resolution of the network-size question)**.** *Let*

$$\delta \coloneqq \frac{3e^{-9/5}}{500}.$$

The following statements hold.

1. *For every integer $k \geq 2$, there is a connected simple stochastic matching instance G_k with $n = 3k$ types such that*

   $$\frac{3}{5n} \leq \lambda_i \leq \frac{9}{5n} \quad \textit{for every type } i,$$

   all rewards belong to the fixed interval $[\delta, 2]$, and

   $$\frac{29e^{-9/5}}{6000}\, n \leq \mathcal{R}^*(G_k) \leq \frac{2+\delta}{3}\, n.$$

   Consequently, $\mathcal{R}^*(G_k) = \Theta(n)$.

   The mean matching linear program of G_k has a unique nondegenerate optimum. In standard form, its minimum positive basic coordinate is

   $$\epsilon = \frac{1}{5k} = \frac{3}{5n}.$$

2. *For every $n \geq 2$, there is a connected n-type instance G for which*

   $$\mathcal{R}^*(G) = 0.$$

   This remains true for an instance whose mean matching linear program has a unique nondegenerate optimum.

Thus no positive function of n can lower-bound $\mathcal{R}^*(G)$ for every n-type instance. On the other hand, the first part shows that a uniform guarantee over connected, bounded-reward, balanced-arrival instances cannot in general be $o(n)$.

3 The connected family

Fix $k \geq 2$ and set $n = 3k$. For every $j \in \{1, \ldots, k\}$, introduce three types

$$c_j, \qquad \ell_j, \qquad h_j,$$

with arrival probabilities

$$\lambda_{c_j} = \frac{3}{5k}, \qquad \lambda_{\ell_j} = \lambda_{h_j} = \frac{1}{5k}.$$

The total probability within each gadget is $1/k$, so these probabilities sum to one over all k gadgets.

Inside gadget j, add the two local edges

$$c_j\ell_j \quad \text{with reward } 1, \qquad c_jh_j \quad \text{with reward } 2.$$

Connect consecutive gadgets by the edges

$$\ell_j\ell_{j+1}, \qquad j = 1, \ldots, k-1,$$

each with reward

$$\delta = \frac{3e^{-9/5}}{500}.$$

Because

$$0 < \delta < 1,$$

all rewards belong to $[\delta, 2]$. The connector edges form a path through the gadgets, and every h_j is joined to that path through c_j and ℓ_j. Hence G_k is connected and simple.

Since $n = 3k$,

$$\lambda_{\ell_j} = \lambda_{h_j} = \frac{1}{5k} = \frac{3}{5n}, \qquad \lambda_{c_j} = \frac{3}{5k} = \frac{9}{5n}.$$

Thus all arrival probabilities lie in the asserted balanced interval.

4 The mean matching linear program

For completeness, consider the mean matching linear program (mean LP)

$$\begin{aligned} \text{maximize} \quad & \sum_{e \in E} r_e y_e, \\ \text{subject to} \quad & \sum_{e \ni i} y_e \le \lambda_i \quad \text{for every } i \in V, \\ & y_e \ge 0 \qquad\quad \text{for every } e \in E. \end{aligned} \tag{LP}$$

Set

$$u \coloneqq \frac{1}{5k},$$

and consider the feasible solution

$$y_{c_j\ell_j} = u, \qquad y_{c_jh_j} = u \quad (j = 1, \ldots, k),$$

with every connector variable equal to zero. Each low-leaf and high-leaf constraint is tight. The load on c_j is $2u$, while its capacity is $3u$, so its slack is u.

Define nonnegative type prices by

$$p_{c_j} = 0, \qquad p_{\ell_j} = 1, \qquad p_{h_j} = 2.$$

For every local edge,

$$p_{c_j} + p_{\ell_j} = 1 = r_{c_j\ell_j}, \qquad p_{c_j} + p_{h_j} = 2 = r_{c_jh_j},$$

and for every connector,

$$p_{\ell_j} + p_{\ell_{j+1}} = 2 > \delta.$$

Thus $p_i + p_{i'} \ge r_{ii'}$ for every edge ii'.

For any feasible vector y, the following identity holds:

$$\begin{aligned} \sum_{i \in V} \lambda_i p_i - \sum_{e \in E} r_e y_e = \sum_{i \in V} p_i \left(\lambda_i - \sum_{e \ni i} y_e \right) \\ + \sum_{e = ii' \in E} \big(p_i + p_{i'} - r_e\big) y_e. \end{aligned} \tag{1}$$

Every term on the right-hand side is nonnegative. The displayed feasible solution makes the left-hand side zero, because its value and the price value are both

$$\sum_{j=1}^{k}(u+2u)=3ku.$$

It is therefore optimal.

Identity (1) also proves uniqueness without invoking any unstated linear-programming result. If y is any other optimal solution, then every nonnegative term on the right-hand side of (1) must vanish. The strict inequality

$$p_{\ell_j}+p_{\ell_{j+1}}-\delta=2-\delta>0$$

forces every connector variable to be zero. Since $p_{\ell_j}=1>0$ and $p_{h_j}=2>0$, the constraints of ℓ_j and h_j must be tight. With all connectors zero, this gives

$$y_{c_j\ell_j}=u, \qquad y_{c_jh_j}=u$$

for every j. Hence the optimum is unique.

To verify nondegeneracy, introduce a slack variable s_i in every type constraint. At the unique optimum, the positive variables are precisely

$$\{y_{c_j\ell_j}, y_{c_jh_j}, s_{c_j} : 1\le j\le k\},$$

and each of these $3k$ variables equals u. For a fixed gadget, the columns of these three variables in the rows (c_j,ℓ_j,h_j) form the matrix

$$\begin{pmatrix}1&1&1\\1&0&0\\0&1&0\end{pmatrix},$$

whose determinant is 1. Across gadgets, these columns form a block-diagonal $3k\times 3k$ matrix and are therefore linearly independent. They constitute a basis, and all associated basic variables are strictly positive. The optimal basic solution is consequently nondegenerate, with

$$\epsilon \coloneqq \min\{y_{c_j\ell_j}, y_{c_jh_j}, s_{c_j} : 1\le j\le k\}=u=\frac{1}{5k}=\frac{3}{5n}.$$

5 A two-horizon linear lower bound

Fix an arbitrary nonanticipative, possibly randomized, anytime policy D. For gadget j, let $O_j(t)$ denote the maximum reward obtainable at time t using only the two local edges

$$c_j\ell_j, \qquad c_jh_j.$$

Let $W_j(t)$ denote the reward that D has earned on these two local edges by time t, and let $Z_D(t)$ be the number of connector matches made by D by time t.

The union of the k local offline-optimal matchings is feasible for the full offline problem, because different gadgets have disjoint type sets. Moreover,

$$W_D(t)=\sum_{j=1}^{k}W_j(t)+\delta Z_D(t).$$

It follows pathwise that

$$\mathrm{OPT}(t) - W_D(t) \geq \sum_{j=1}^{k} \bigl(O_j(t) - W_j(t)\bigr) - \delta Z_D(t). \tag{2}$$

Each local matching selected by D is feasible for the corresponding local offline problem, so every difference $O_j(t) - W_j(t)$ is nonnegative.

Set the first horizon equal to

$$T = k.$$

For gadget j, let E_j be the event that periods $1, \ldots, k$ contain exactly one arrival of type c_j, exactly one arrival of type ℓ_j, and no arrival of type h_j; arrivals of all other types are unrestricted. After choosing the distinct positions of the c_j- and ℓ_j-arrivals, every remaining arrival must avoid all three types of gadget j, whose total probability is $1/k$. Therefore

$$\begin{aligned} p_k \coloneqq \mathbb{P}(E_j) &= k(k-1) \cdot \frac{3}{5k} \cdot \frac{1}{5k} \left(1 - \frac{1}{k}\right)^{k-2} \\ &= \frac{3}{25} \left(1 - \frac{1}{k}\right)^{k-1}. \end{aligned} \tag{3}$$

Let X_j be the indicator that D has executed the local match $c_j \ell_j$ by time k, and define

$$x \coloneqq \sum_{j=1}^{k} \mathbb{P}\bigl(E_j \cap \{X_j = 1\}\bigr). \tag{4}$$

On $E_j \cap \{X_j = 0\}$, the local offline optimum matches the unique c_j to the unique ℓ_j and earns 1, whereas D has earned no local reward in gadget j. Hence

$$O_j(k) - W_j(k) \geq 1 \quad \text{on } E_j \cap \{X_j = 0\}.$$

Every match consumes two agents, so

$$Z_D(k) \leq \frac{k}{2}.$$

Taking expectations in (2) and using (3)–(4) gives

$$\begin{aligned} \mathrm{Reg}_D(k) &\geq \sum_{j=1}^{k} \mathbb{P}\bigl(E_j \cap \{X_j = 0\}\bigr) - \frac{\delta k}{2} \\ &= k p_k - x - \frac{\delta k}{2}. \end{aligned} \tag{5}$$

For the second block of periods $k+1, \ldots, 2k$, let F_j be the event that this block contains exactly one arrival of type h_j and no arrival of type c_j; arrivals of type ℓ_j and of types outside gadget j are unrestricted. Choosing the position of the unique h_j-arrival and requiring every other position to avoid c_j and h_j gives

$$\begin{aligned} q_k \coloneqq \mathbb{P}(F_j) &= k \cdot \frac{1}{5k} \left(1 - \frac{4}{5k}\right)^{k-1} \\ &= \frac{1}{5} \left(1 - \frac{4}{5k}\right)^{k-1}. \end{aligned} \tag{6}$$

The event F_j depends only on the fresh arrivals in periods $k+1, \dots, 2k$. It is therefore independent of the arrivals, policy decisions, and policy randomization through time k. In particular,

$$\mathbb{P}\big(E_j \cap \{X_j = 1\} \cap F_j\big) = q_k \, \mathbb{P}\big(E_j \cap \{X_j = 1\}\big). \tag{7}$$

On $E_j \cap \{X_j = 1\} \cap F_j$, exactly one c_j and exactly one h_j arrive over the entire $2k$-period horizon. The policy has already used that unique center with an ℓ_j-agent for reward 1, whereas the local offline optimum instead matches the center to the high leaf for reward 2. There is no second center available to the policy. Consequently,

$$O_j(2k) - W_j(2k) = 1 \quad \text{on } E_j \cap \{X_j = 1\} \cap F_j. \tag{8}$$

No independence between different gadgets is needed. Since at most k matches of any kind can be made from $2k$ arrivals,

$$Z_D(2k) \le k.$$

Equations (2), (7), and (8) therefore imply

$$\mathrm{Reg}_D(2k) \ge q_k x - \delta k. \tag{9}$$

Multiplying (5) by q_k and adding (9) cancels the policy-dependent quantity x:

$$q_k \, \mathrm{Reg}_D(k) + \mathrm{Reg}_D(2k) \ge k \left[p_k q_k - \delta \left(1 + \frac{q_k}{2}\right)\right]. \tag{10}$$

If $\sup_t \mathrm{Reg}_D(t) = +\infty$, the desired lower bound is immediate. Otherwise, both regrets on the left-hand side of (10) are at most $\sup_t \mathrm{Reg}_D(t)$, and hence

$$\sup_{t \in \mathbb{N}} \mathrm{Reg}_D(t) \ge k \, \frac{p_k q_k - \delta(1 + q_k/2)}{1 + q_k}. \tag{11}$$

We now bound all constants uniformly for $k \ge 2$. The elementary inequality

$$\log(1-y) \ge -\frac{y}{1-y}, \qquad 0 < y < 1, \tag{12}$$

follows because

$$\frac{d}{dy}\left(\log(1-y) + \frac{y}{1-y}\right) = \frac{y}{(1-y)^2} \ge 0$$

and the expression vanishes at $y = 0$.

Taking $y = 1/k$ in (12) gives

$$(k-1)\log\left(1 - \frac{1}{k}\right) \ge -1,$$

and therefore

$$p_k \ge p_0 \coloneqq \frac{3}{25e}. \tag{13}$$

Taking $y = 4/(5k)$ gives

$$\begin{aligned}(k-1)\log\left(1 - \frac{4}{5k}\right) &\ge -\frac{4(k-1)}{5k-4} \\ &\ge -\frac{4}{5},\end{aligned}$$

where the last inequality follows from

$$\frac{4(k-1)}{5k-4} \leq \frac{4}{5}.$$

Thus

$$q_k \geq q_0 \coloneqq \frac{e^{-4/5}}{5}. \tag{14}$$

The definition (6) also gives

$$q_k \leq \frac{1}{5}. \tag{15}$$

The chosen connector reward satisfies

$$\delta = \frac{3e^{-9/5}}{500} = \frac{1}{4}\, p_0 q_0. \tag{16}$$

Using (13)–(16), the numerator in (11) obeys

$$\begin{aligned} p_k q_k - \delta\left(1 + \frac{q_k}{2}\right) &\geq p_0 q_0 - \frac{p_0 q_0}{4}\left(1 + \frac{1}{10}\right) \\ &= \frac{29}{40}\, p_0 q_0. \end{aligned} \tag{17}$$

Furthermore, $1 + q_k \leq 6/5$, so

$$\frac{p_k q_k - \delta(1 + q_k/2)}{1 + q_k} \geq \frac{29}{48}\, p_0 q_0. \tag{18}$$

Substituting

$$p_0 q_0 = \frac{3e^{-9/5}}{125}$$

into (18) and combining with (11) yields

$$\sup_{t \in \mathbb{N}} \operatorname{Reg}_D(t) \geq \frac{29e^{-9/5}}{2000}\, k. \tag{19}$$

This holds for every nonanticipative anytime policy D, including every randomized policy. Taking the infimum over D and using $n = 3k$ gives

$$\mathcal{R}^*(G_k) \geq \frac{29e^{-9/5}}{6000}\, n. \tag{20}$$

The use of two horizons is essential to the argument. A policy that retains centers in anticipation of high leaves loses at horizon k, while a policy that commits centers to low leaves makes a constant fraction of irrevocably incorrect commitments when evaluated at horizon $2k$. The same anytime policy is evaluated at both horizons.

6 A uniform linear upper bound

Consider the following anytime policy D_{loc}. It never uses connector edges. Within each gadget, it matches whenever an unmatched center and an unmatched leaf coexist, giving priority to an available high leaf h_j over a low leaf ℓ_j.

Fix a horizon and a realized arrival sequence in one gadget. Let

$$C, \qquad L, \qquad H$$

be the numbers of center, low-leaf, and high-leaf arrivals, respectively, and let D_L and D_H be the numbers of low and high local matches made by the policy.

Because the policy is work-conserving within the gadget, after it acts there can never be both an unmatched center and an unmatched leaf. It follows that

$$D_L + D_H = \min(C, L + H). \tag{21}$$

The local offline optimum receives one unit for every match and one additional unit for every high match. It can make $\min(C, L + H)$ total matches and $\min(C, H)$ high matches simultaneously: it first matches $\min(C, H)$ high leaves and then uses any remaining centers on low leaves. Hence its value is

$$O = \min(C, L + H) + \min(C, H). \tag{22}$$

The policy's local value is

$$W = D_L + 2D_H = \min(C, L + H) + D_H. \tag{23}$$

Consequently,

$$O - W = \min(C, H) - D_H. \tag{24}$$

Define

$$Q \coloneqq (L + H - C)^+, \qquad a^+ \coloneqq \max\{a, 0\}.$$

We verify

$$O - W \leq Q \tag{25}$$

in every possible count regime.

If $C \geq L + H$, then every leaf is matched, so $D_H = H$. Thus

$$O - W = 0 = Q.$$

If

$$H < C < L + H,$$

then (21) gives $D_L + D_H = C$. Since $D_L \leq L$,

$$D_H \geq C - L.$$

In this regime $\min(C, H) = H$, and therefore

$$O - W = H - D_H \leq H - (C - L) = L + H - C = Q.$$

Finally, if $C \leq H$, then (21) gives $D_L + D_H = C$, and hence

$$O - W = C - D_H = D_L \leq L.$$

Since $H \geq C$,

$$L \leq L + H - C = Q.$$

This completes the proof of (25), including all boundary cases.

We next bound $\mathbb{E}Q$ uniformly over the horizon. For a fixed gadget, let

$$S_t = L_t + H_t - C_t,$$

where the counts are taken through time t. Its increments are independent, with distribution

$$\Delta S_t = \begin{cases} +1, & \text{with probability } \dfrac{2}{5k}, \\ -1, & \text{with probability } \dfrac{3}{5k}, \\ 0, & \text{with probability } 1 - \dfrac{1}{k}. \end{cases} \tag{26}$$

Take

$$\theta = \log\left(\frac{3}{2}\right) > 0.$$

A direct calculation gives

$$\begin{aligned} \mathbb{E}e^{\theta \Delta S_t} &= 1 - \frac{1}{k} + \frac{2}{5k} \cdot \frac{3}{2} + \frac{3}{5k} \cdot \frac{2}{3} \\ &= 1. \end{aligned} \tag{27}$$

Since $S_0 = 0$ and the increments are independent,

$$\mathbb{E}e^{\theta S_t} = 1 \quad \text{for every } t \in \mathbb{N}. \tag{28}$$

Markov's inequality applied to the nonnegative integrable random variable $e^{\theta S_t}$ yields, for every integer $r \geq 1$,

$$\mathbb{P}(S_t \geq r) = \mathbb{P}\big(e^{\theta S_t} \geq e^{\theta r}\big) \leq e^{-\theta r}. \tag{29}$$

Because S_t^+ is a nonnegative integer-valued random variable, its tail-sum representation, justified by monotone convergence, gives

$$\begin{aligned} \mathbb{E}S_t^+ &= \sum_{r=1}^{\infty} \mathbb{P}(S_t \geq r) \\ &\leq \sum_{r=1}^{\infty} e^{-\theta r} = \sum_{r=1}^{\infty} \left(\frac{2}{3}\right)^r = 2. \end{aligned} \tag{30}$$

Thus, for every gadget j, every horizon t, and

$$Q_j(t) \coloneqq (L_j(t) + H_j(t) - C_j(t))^+,$$

we have

$$\mathbb{E}Q_j(t) \leq 2. \tag{31}$$

It remains to account for connector edges in the full offline optimum. Fix a realized arrival multiset and an offline-optimal matching, and let z_j be the number of ℓ_j-agents used as endpoints of connector matches. We claim that

$$z_j \leq Q_j \tag{32}$$

for every j.

Suppose instead that $z_j > Q_j$. Since

$$Q_j = (L_j + H_j - C_j)^+,$$

this strict inequality implies

$$z_j > L_j + H_j - C_j,$$

and hence

$$C_j > L_j + H_j - z_j. \tag{33}$$

After the z_j low-leaf agents assigned to connectors are removed, at most $L_j + H_j - z_j$ leaf agents remain available for local matches with c_j-agents. Centers have no incident connector edges. By (33), at least one c_j-agent is therefore unmatched in the offline matching.

Because $z_j > 0$, some connector match uses an ℓ_j-agent. Remove that connector match, freeing its ℓ_j-endpoint, and match the freed ℓ_j-agent to an unmatched c_j-agent. This operation remains feasible, loses connector reward δ, and gains local reward 1. Its net reward change is

$$1 - \delta > 0,$$

contradicting optimality. This proves (32).

Every connector match has two low-leaf endpoints. Therefore, if Z^* denotes the number of connector matches in the offline optimum,

$$2Z^* = \sum_{j=1}^{k} z_j \leq \sum_{j=1}^{k} Q_j. \tag{34}$$

The offline connector reward is consequently at most

$$\delta Z^* \leq \frac{\delta}{2} \sum_{j=1}^{k} Q_j. \tag{35}$$

In each gadget, the local part of any full offline matching has value at most the standalone local optimum O_j. Combining this observation with (35) gives the pathwise upper bound

$$\mathrm{OPT}(t) \leq \sum_{j=1}^{k} O_j(t) + \frac{\delta}{2} \sum_{j=1}^{k} Q_j(t). \tag{36}$$

The policy D_{loc} uses no connectors, so its total reward is

$$W_{D_{\mathrm{loc}}}(t) = \sum_{j=1}^{k} W_j(t).$$

Using (25) in (36), we obtain pathwise

$$\mathrm{OPT}(t) - W_{D_{\mathrm{loc}}}(t) \leq \left(1 + \frac{\delta}{2}\right) \sum_{j=1}^{k} Q_j(t). \tag{37}$$

Taking expectations and applying (31) yields, uniformly in t,

$$\begin{aligned}\mathrm{Reg}_{D_{\mathrm{loc}}}(t) &\le \left(1+\frac{\delta}{2}\right)\sum_{j=1}^{k}\mathbb{E}Q_j(t)\\ &\le \left(1+\frac{\delta}{2}\right)2k\\ &= (2+\delta)k.\end{aligned}\tag{38}$$

Hence

$$\mathcal{R}^*(G_k) \le \sup_{t\in\mathbb{N}} \mathrm{Reg}_{D_{\mathrm{loc}}}(t) \le (2+\delta)k = \frac{2+\delta}{3}\,n.\tag{39}$$

Together, (20) and (39) prove

$$\boxed{\frac{29e^{-9/5}}{6000}\,n \le \mathcal{R}^*(G_k) \le \frac{2+\delta}{3}\,n}.$$

7 Zero regret for arbitrarily large connected instances

Fix any $n \ge 2$. Let G be a star with center c, leaves $\ell_1, \dots, \ell_{n-1}$, and unit reward on every edge. For definiteness, choose

$$\lambda_c = \frac{3}{4}, \qquad \lambda_{\ell_i} = \frac{1}{4(n-1)} \quad (1 \le i \le n-1).$$

All probabilities are positive and sum to one, and the graph is connected.

Consider the policy that matches a center and any leaf whenever they coexist. Let

$$C_t \qquad \text{and} \qquad L_t \coloneqq \sum_{i=1}^{n-1} A_{\ell_i}(t)$$

be the total center and leaf arrivals through time t, where $A_{\ell_i}(t)$ denotes the number of type-ℓ_i arrivals through time t, and let M_t be the number of matches made by the policy. After the policy acts, an unmatched center and an unmatched leaf cannot coexist. Since the numbers of unmatched centers and leaves are respectively

$$C_t - M_t \qquad \text{and} \qquad L_t - M_t,$$

at least one of these quantities is zero. It follows that

$$M_t = \min(C_t, L_t).\tag{40}$$

No matching can use more than C_t centers or more than L_t leaves, so the offline optimum also has exactly

$$\min(C_t, L_t)$$

unit-reward matches. Thus

$$W_D(t) = \mathrm{OPT}(t)$$

on every sample path and at every horizon. Consequently,

$$\mathrm{Reg}_D(t) = 0 \quad \text{for every } t, \qquad \mathcal{R}^*(G) = 0. \tag{41}$$

This star also has a unique nondegenerate mean-LP optimum. Every leaf has only one incident edge, the aggregate leaf rate is 1/4, and the center rate is 3/4. The unique optimum is

$$y_{c\ell_i} = \lambda_{\ell_i} = \frac{1}{4(n-1)} \quad (1 \le i \le n-1),$$

with center slack

$$s_c = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}.$$

The $n-1$ positive edge variables together with s_c give n positive variables. Their columns are linearly independent because each edge column is the only one among them with a nonzero coordinate in its corresponding leaf row, while the center-slack column is supported only in the center row. Hence the optimum is nondegenerate.

Therefore, even among connected instances satisfying a unique nondegenerate mean-LP condition, network size does not force regret to increase on every instance.

8 Reward scaling

There is also no reward-normalization-independent lower bound depending only on n.

Proposition 1. *Fix a compatibility graph M, an arrival distribution λ, a reward vector r, and a scalar $a > 0$. Then*

$$\mathcal{R}^*(M, \lambda, ar) = a\,\mathcal{R}^*(M, \lambda, r).$$

Proof. For every realized arrival sequence and every feasible matching, replacing r by ar multiplies the matching's reward by a. Hence

$$\mathrm{OPT}_{ar}(t) = a\ \mathrm{OPT}_r(t).$$

The set of nonanticipative history-to-action rules is unchanged by this scaling. Pair each policy under r with the identical history-to-action rule under ar. For such corresponding policies,

$$W_{D,ar}(t) = a\,W_{D,r}(t)$$

pathwise, and therefore

$$\mathrm{Reg}_{D,ar}(t) = a\ \mathrm{Reg}_{D,r}(t).$$

Since $a > 0$, taking the supremum over horizons and then the infimum over the identical policy spaces gives

$$\begin{aligned}\mathcal{R}^*(M, \lambda, ar) &= \inf_D \sup_t a\ \mathrm{Reg}_{D,r}(t) \\ &= a \inf_D \sup_t \mathrm{Reg}_{D,r}(t) \\ &= a\,\mathcal{R}^*(M, \lambda, r).\end{aligned}$$

□

9 Conclusion and scope

The result is a complete solution to the stated question about raw dependence on the number of types. The two relevant quantifiers have different answers.

1. There is no positive increasing function of n that lower-bounds $\mathcal{R}^*(G)$ for every n-type instance: connected unit-reward stars have zero regret for arbitrarily large n.

2. Worst-case regret can grow linearly with n. The connected family G_k has fixed bounded rewards, arrival probabilities of order $1/n$, a unique nondegenerate mean-LP optimum, and
$$\mathcal{R}^*(G_k) = \Theta(n).$$
Thus a uniform raw regret guarantee over any class containing these instances cannot be $o(n)$.

For the constructed family,
$$\epsilon = \frac{3}{5n},$$
so the proved behavior may equivalently be written as
$$\mathcal{R}^*(G_k) = \Theta(n) = \Theta(\epsilon^{-1}).$$
No claim stronger than this is established. In particular, the argument does not give
$$\mathcal{R}^*(G_k) = \Omega(n/\epsilon),$$
which would be quadratic in n for this family, and it does not prove that an explicit multiplicative n in an $O(n/\epsilon)$ upper bound is tight. That stronger issue lies outside the scope of the network-size question resolved here.

References

[1] Süleyman Kerimov, Itai Ashlagi, Itai Gurvich. *On the Optimality of Greedy Policies in Dynamic Matching*. Economics and Computation (EC), 2022. https://doi.org/10.1145/3490486.3538323